

CNN Results for Young's Modulus Estimation

Lamb-wave simulation data | Pristine 1 mm plate | 120 kHz | 15 dB noise

Training Set-up (CNN)

- Training dataset: 30 known Young's modulus cases
- Input: Lamb-wave waveform nearest the 75 mm sensor location
- Output: Predicted Young's modulus in GPa
- Training epochs: 1,500
- Final reported training RMSE: 4.302 GPa
- Test files: 43.02 GPa, 50 GPa, and 91 GPa

Unknown Test Results

Inverse PINN Results on Unknown Moduli

Unknown E	CNN predicted E	Absolute error	Error percentage
43.02 GPa	50.3909 GPa	7.3709 GPa	17.13%
50.00 GPa	48.8689 GPa	1.1311 GPa	2.26%
91.00 GPa	66.8907 GPa	24.1093 GPa	26.49%

Observations

- The CNN estimated the 50 GPa waveform most accurately, with 2.26% error.
- The CNN overestimated 43.02 GPa and underestimated 91 GPa.
- The average error across the three unseen cases was 15.29%.
- The model's accuracy was inconsistent across the modulus range, likely because the training dataset contained only 30 simulated cases.

PINN vs. CNN

Model	50 GPa error	91 GPa error
PINN	2.05%	43.93%
One-sensor CNN	2.26%	26.49%

Both models estimated 50 GPa accurately. The CNN produced a lower error than the PINN for the 91 GPa test.